Mathematical Proof (W.I.P)

The Abel-Ruffini theorem disallows the use of algebraic operations and radicals. Hence, it can be solved using higher transcendental functions in closed form, which were proven by notable mathematicians that elliptic modular and Jacobi theta functions functions provide a framework for a solution.

1) Reduction Framework:

The reduction framework or as— Tschirnhaus transformation.

To find an expression for the roots of a general quintic equation. For example—

$$x^5 + a_4x^4 + a_3x^3 + a_2x^2 + a_1x + a_0 = 0$$

But since, you can’t directly apply theta functions to the full polynomial.

So, you must first simplify it to a canonical normal form— such as the Bring-Jerrad form. Thus, we get the following—

$$y^5 + y + c = 0$$

Which is done through a Tschirnhais transformation to eliminate the higher-order terms. Thus, the complexity of the original polynomial is concentrated entirely into the single invariant coefficient c.

2) Constructing Solutions:

Once reduced to the Bring-Jerrard or principal quintic form, the roots can be expressed using modular functions and Jacobi theta functions ($\vartheta_1, \vartheta_2, \vartheta_3$)

Hence —

- The coefficient c is mapped to the elliptic modular lambda function $\lambda(\tau)$ or the absolute modular invariant $j(\tau)$ over the upper half-plane
- The roots are then formulated using quotients of theta functions evaluated at the appropriate nome τ, capturing the icosahedral symmetry of the quintic's Galois group (A_5)

For example, taking $c = 1$—

$$y^5 + y + 1 = 0$$

This equation has one real root and four complex conjugate roots. So, for this method, we'll focus on calculating the real root analytically using theta functions(yes i am talking to YOU TUKUN)

Thus, to solve $y^5 + y + c = 0$ using Jacobi theta functions, we use the connection established by Charles Hermite and Francesco Brioschi between the icosahedral symmetry group A_5 and modular functions.

Hence, we must calculate the invariant parameter of u

The parameter $c = 1$ is related to a modular parameter u via the fundamental algebraic relation, which is—

$$c = \frac{1 - u^2}{2u^{1/2}(1 + u^2)^{3/4}}$$

Squaring and rearranging gives a rational equation in u, which can be solved for $u \in (0, 1)$. For $c = 1$, $u \approx 0.1306$.

Using u, we define the elliptic modulus $k = u^2 \approx 0.01706$. Then, the complete elliptic integral of the first kind $K(k)$ and its complementary integral $K'(k) = K(\sqrt{1 - k^2})$ gives the **nome q. Hence—**

$$q = \exp\left(-\pi \frac{K'(k)}{K(k)}\right)$$

Now, evaluating the roots via the Jacobi Theta Functions. Since, the Jacobi theta functions $\vartheta_2(q), \vartheta_3(q), \vartheta_4(q)$ are defined as fast-converging infinite series—

$$\vartheta_2(q) = 2q^{1/4} \sum_{n=0}^{\infty} q^{n(n+1)} = 2q^{1/4}(1 + q^2 + q^6 + q^{12} + \dots)$$

$$\vartheta_3(q) = 1 + 2 \sum_{n=1}^{\infty} q^{n^2} = 1 + 2q + 2q^4 + 2q^9 + \dots$$

$$\vartheta_4(q) = 1 + 2\sum_{n=1}^{\infty}(-1)^n q^{n^2} = 1 - 2q + 2q^4 - 2q^9 + \dots$$

The real root y_0 of $y^5 + y + 1 = 0$ can be expressed via the quotient of theta functions evaluated at q and $q^{1/5}$ or—

$$y_0 = -\frac{\vartheta_2(q^{1/5}) \cdot \vartheta_3(q^{1/5})}{\vartheta_2(q) \cdot \vartheta_3(q)} \cdot \Phi(q)$$

And which $\Phi(q)$ involves ratios of theta function evaluations capturing the 5-fold modular transformation, hence it can be expressed as—

$$\Phi(q) = \frac{\vartheta_4(q^5)}{\vartheta_4(q)}$$

Since, $q \approx 1.82 \times 10^{-5}$ is extremely small, $q^{1/5} \approx 0.1132$. Hence, the Theta series converges rapidly with just the two or three terms.

Hence, the Theta values are—

- $\vartheta_2(q^{1/5}) \approx 2(0.1132)^{1/4}(1 + 0.1132^2) \approx 1.1788$
- $\vartheta_3(q^{1/5}) \approx 1 + 2(0.1132) + 2(0.1132)^4 \approx 1.2267$
- $\vartheta_2(q) \approx 2(1.82 \times 10^{-5})^{1/4} \approx 0.1306$
- $\vartheta_3(q) \approx 1.0000$

Evaluating the ratio, by substituting these values into the modular expressions yields—

$$y_0 \approx -0.754877669$$

Verifying the solution, let's check if $y^5 + y + 1$ at $y = -0.754877669$—

$$(-0.754877669)^5 + (-0.754877669) + 1 = -0.245122331 - 0.754877669 + 1 = 0$$

3) Optimization:

The optimization or efficiency can be easily summed by the following—

- While theta-function expressions provide exact, closed-form analytic solutions, evaluating nome values and high-precision modular functions symbolically or

numerically can introduce significant computational overhead compared to simpler algebraic forms.

- For computational efficiency in software, iterative numerical root-finding algorithms — such as the **Durand-Kerner** or **Aberth-Ehrlich** methods are superior than the closed-form transcendental expansions.